



Sri Chaitanya IIT Academy.,India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & IIT

JEE-MAIN

Date: 17-12-2025

Time: 09.00Am to 12.00Pm

GTM-01

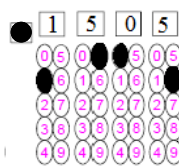
Max. Marks: 300

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
 2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
 3. The test is of **3 hours** duration.
 4. The Test Booklet consists of **75 Questions**. The maximum marks are **300**.
 5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics and Chemistry** having **25 Questions** in each subject and subject having **two sections**.
(I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
(II) Section-II contains **05 Numerical Value Type Questions**.
- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking Question Cancelled for Marking

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: _____

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17-12-2025_Sr. XL-500, Sr.S60(N&S) & LIIT Jee-Main_GTM-01_Test Syllabus

MATHEMATICS : On ONLY SECOND Year SYLLABUS

PHYSICS : On ONLY SECOND Year SYLLABUS

CHEMISTRY : On ONLY SECOND Year SYLLABUS

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**MATHEMATICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

01. If $y = f(x)$ satisfies the differential equation $\sin x \frac{dy}{dx} + 2y \cos x = 8$ with $f\left(\frac{\pi}{2}\right) = 8$ then the value of $\frac{3}{4}f\left(\frac{\pi}{3}\right)$ is _____
- 1) 16 2) 4 3) 6 4) 8
02. Let $x_1, x_2, x_3, \dots, x_{10}$ be ten observations such that $\sum_{i=1}^{10} (x_i - 2) = 30$, $\sum_{i=1}^{10} (x_i - \beta)^2 = 98$, $\beta < 8$ and their variance is $\frac{4}{5}$. If μ and σ^2 are respectively the mean and the variance of $2(x_1 - 5) + 7\beta, 2(x_2 - 5) + 7\beta, \dots, 2(x_{10} - 5) + 7\beta$ then $\frac{4\beta\mu}{\sigma^2}$ is equal to
- 1) 90 2) $\frac{35}{4}$ 3) 35 4) 100
03. If $\left({}^{100}C_1\right)^2 + 2\left({}^{100}C_2\right)^2 + 3\left({}^{100}C_3\right)^2 + \dots + 100\left({}^{100}C_{100}\right)^2 = \frac{N(200!)}{(100!)^2}$ then the sum of the positive integral divisors of N^2 is _____
- 1) 5647 2) 5476 3) 5764 4) 5467
04. The equations of tangents at two points P, Q on a parabola are $3x + 4y - 7 = 0$, $2x + 3y - 10 = 0$. If $x - y = 0$ is the tangent at vertex A of the same parabola then the length of the latus rectum of the parabola is
- 1) $\sqrt{2}$ 2) $4\sqrt{2}$ 3) $2\sqrt{2}$ 4) $8\sqrt{2}$
05. Let $f(x)$ be a continuous function in $(0,1)$ satisfying $\int_0^1 x\sqrt{x}f(x)(1-\sqrt{x}f(x))dx = \frac{1}{8}$. Then the number of solutions of the equation $f(x) = e^x$ is
- 1) 1 2) 0 3) 2 4) 3

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06. Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. Then the number of onto functions f defined from A to A such that $f(i) = i$ for at least four values of i from $i = 1, 2, 3, 4, 5, 6, 7$ is
- 1) 92 2) 7! 3) 260 4) 126
07. Let S and S^1 be the foci of an ellipse $\frac{x^2}{50} + \frac{y^2}{25} = 1$ and P be a point on the ellipse. If α, β are the minimum and maximum values of $(SP)(S^1P)$ respectively then $\beta + 2\alpha$ value is _____
- 1) 0 2) 10 3) 125 4) 100
08. Let $\int (x^2 - 1)e^x (x^2 + 4x + 1) dx = e^x f(x) + C$, C -constant of integration. If $g(x) = f(x) + f^1(x)$ then the absolute difference of non integral roots of $g(x) = 0$ is _____
- 1) $3\sqrt{2}$ 2) 0 3) 2 4) $2\sqrt{3}$
09. Matrices of order 2×2 are formed by using the elements of the set $A = \{-2, -1, 0, 1, 2\}$, then the probability that the matrix is symmetric (or) skew symmetric is
- 1) $\frac{130}{625}$ 2) $\frac{129}{625}$ 3) $\frac{125}{625}$ 4) $\frac{1}{5}$
10. For any complex number $\omega = c + id$, let $\arg(\omega) \in (-\pi, \pi]$ where $i = \sqrt{-1}$. Let α and β be real numbers ($\alpha < \beta$) such that for all complex numbers $z = x + iy$ satisfying $\arg\left(\frac{z + \alpha}{z + \beta}\right) = \frac{\pi}{4}$, the ordered pair (x, y) lies on the circle $x^2 + y^2 + 5x - y + 6 = 0$. Then consider
- Statement-I : $\alpha + 2\beta = 8$
- Statement-II : $\alpha\beta = 4$
- 1) Both Statement-I and Statement-II are correct
- 2) Both Statement-I and Statement-II are incorrect
- 3) Statement-I is correct and Statement-II is incorrect
- 4) Statement-I is incorrect and Statement-II is correct





11. Let $f(x) = ax^2 + bx + c$ ($x \in R, a > 0$) such that $f(2)$ is minimum value of $f(x)$ then
- 1) $f(2) < f(1) < f(3)$
 - 2) $f(2) < f(0) < f(5)$
 - 3) $f(-1) < f(0) < f(1)$
 - 4) $f(2) < f(0) = f(3)$
12. y-intercept of the common tangent to the parabolas $y^2 = 32x$ and $x^2 = 108y$ is
- 1) -18
 - 2) -12
 - 3) -9
 - 4) -6
13. Suppose that $f: R \rightarrow R$ be a continuous function such that $f(3x) = f(x) \forall x \in R$ and $f(2) = 5$. Then the value of $\int_{12}^{17} \text{sgn}(f(x)) dx$ is equal to _____ (Where $\text{sgn}(x)$ represents signum function)
- 1) 5
 - 2) 25
 - 3) 30
 - 4) 16
14. Consider a branch of the hyperbola $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A . Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to point A then the area of the triangle ABC is _____ (in square units)
- 1) $\sqrt{\frac{3}{2}}$
 - 2) $\sqrt{\frac{2}{3}}$
 - 3) $\sqrt{\frac{3}{2}} - 1$
 - 4) $1 - \sqrt{\frac{2}{3}}$
15. Let $f: R \rightarrow R$ be a twice differentiable function such that $f(x+y) = f(x)f(y)$ for all $x, y \in R$. If $f'(0) = 6a$, and f satisfies $f''(x) - 4af'(x) - 12f(x) = 0$, $a < 0$. Then the area of the region $R = \{(x, y) / 0 \leq y \leq f(ax), 0 \leq x \leq 1\}$ is _____
- 1) $\frac{e^6 + 1}{6}$
 - 2) $\frac{e^6 + e}{6}$
 - 3) $\frac{e^6 - e}{6}$
 - 4) $\frac{e^6 - 1}{6}$
16. Number of non-zero solutions of the equation $x + \int_0^x \ln t \, dt = \frac{x^2}{3}$ is equal to
- 1) 0
 - 2) 3
 - 3) 2
 - 4) 1





17. Let $A = [a_{ij}]$ be a 2×2 matrix such that $a_{ij} \in \{0, 1\}$ for all i, j . Let the random variable X denote the possible values of the determinant of the matrix A . Then match the following :

Column-I**Column-II**

A) $P(X = 0)$

1) $\frac{13}{16}$

B) $P(X \geq 0)$

2) 0

C) Mean of X

3) $\frac{5}{8}$

D) Variance of X

4) $\frac{3}{8}$

1) A-2; B-3; C-1; D-4

2) A-3; B-2; C-1; D-4

3) A-2; B-4; C-3; D-1

4) A-3; B-1; C-2; D-4

18. Let $f(x) = 11x^5 + 7x^3 + x + 1$ and $g(x)$ be the inverse function of $f(x)$. Then the area of the region bounded by $y = g(x)$, x-axis and $x = 20$ is

1) $\frac{189}{12}$

2) $\frac{179}{12}$

3) $\frac{59}{12}$

4) $\frac{199}{12}$

19. The centre of a circle which bisects the circumferences of two circles $x^2 + y^2 = 9$ and $x^2 + y^2 - 2x - 4y + 2 = 0$ lies on

1) $2x - 4y + 3 = 0$

2) $2x - 4y - 1 = 0$

3) $2x + 4y + 3 = 0$

4) $2x + 4y + 1 = 0$

20. Let α and β be the roots of $x^2 + \sqrt{3}x - 16 = 0$ and γ and δ be the roots of $x^2 + 3x - 1 = 0$. If $P_n = \alpha^n + \beta^n$, $Q_n = \gamma^n + \delta^n$ ($n \in N$) then

$$3 \left(\frac{P_{26} + \sqrt{3}P_{25}}{8P_{24}} \right) - 6 \left(\frac{Q_{26} - Q_{24}}{Q_{25}} \right) = \text{---}$$

1) 24

2) 18

3) 17

4) 6

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SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Let $A(-4,0)$ and $B(4,0)$. Then the number of points $C = (x,y)$ on the circle $x^2 + y^2 = 16$ such that the area of the triangle whose vertices are A, B and C is a positive integer is _____
22. Let $y = f(x)$ be an invertible function such that x -intercept of the tangent at any point $P(x,y)$ on $y = f(x)$ is equal to the square of abscissae of the point of tangency. If $f(2) = 1$ then $2f^{-1}\left(\frac{5}{8}\right) =$ _____
23. The number of words which can be formed using all the letters of the word 'MIMIXA' in which no two alike letters are together is _____
24. If $\int_0^{\pi/2} \sin^{10} x \cos^{10} x \, dx = 2^{-n} \int_0^{\pi/2} (\sin^{10} x + \cos^{10} x) \, dx (n \in N)$ then the value of $2n$ is equal to _____
25. If $a = {}^{100}C_0 + {}^{100}C_3 + {}^{100}C_6 + \dots + {}^{100}C_{99}$, $b = {}^{100}C_1 + {}^{100}C_2 + {}^{100}C_4 + {}^{100}C_5 + \dots$, $c = {}^{100}C_1 - {}^{100}C_2 + {}^{100}C_4 - {}^{100}C_5 + \dots$ then thrice the value of $\left(a - \frac{b}{2}\right)^2 + \frac{3}{4}c^2$ is _____

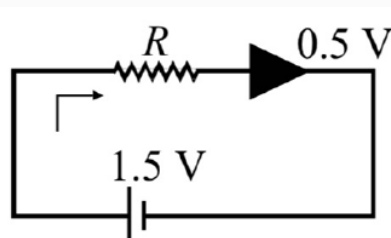


**PHYSICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. The diode used in the circuit shown in the figure has constant voltage drop of 0.5 V at all currents and a maximum power rating of 100 milli watt. What should be the value of the resistor R, connected in series with diode for obtaining maximum current?

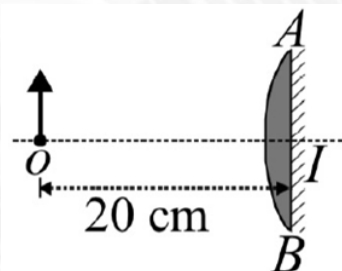


- 1) 6.76Ω 2) 20Ω 3) 5Ω 4) 5.6Ω
27. Following diagram performs the logic function of
-
- 1) AND gate 2) NAND gate 3) OR gate 4) XOR gate
28. Assertion (A): Density of all the nuclei is same.
Reason (R): Radius of nucleus is directly proportional to the cube root of mass number
- 1) Both (A) and (R) are correct and (R) is correct explanation of (R)
2) Both (A) and (R) are correct and (R) is not correct explanation of (R)
3) (A) is true (R) is false
4) (A) is false (R) is true
29. A photosensitive metallic surface has work function, $h\nu_0$. If photons of energy $2h\nu_0$ fall on this surface, the electrons come out with a maximum speed of $4 \times 10^6 \text{ m/s}$. When the photon energy is increased to $5h\nu_0$, then maximum speed of photoelectrons will be
- 1) $2 \times 10^7 \text{ m/s}$ 2) $2 \times 10^6 \text{ m/s}$ 3) $8 \times 10^6 \text{ m/s}$ 4) $8 \times 10^5 \text{ m/s}$

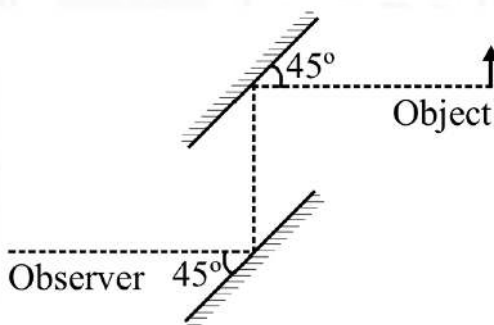
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30. Using Moesley's law, i.e, $\sqrt{\nu} = a(Z - b)$ the wavelength of the K_{α} line, for Z equal to 41 when $a = 5 \times 10^7 (Hz)^{1/2}$, is
- 1) $2 \times 10^8 m$ 2) $4 \times 10^{18} m$ 3) $7.5 \times 10^{-11} m$ 4) $0.75 \times 10^{-8} m$
31. A beam of light composed of red and green rays is incident obliquely at a point on the face of rectangular glass slab. When coming out on the opposite parallel face, the red and green rays emerge from
- 1) Two points propagating in two different non parallel directions.
2) Two points propagating in same direction
3) One point propagating in two different directions.
4) One point propagating in the same direction.
32. A point object is placed at a distance of 20 cm from a thin plano-convex lens of focal length 15 cm. If the plane surface is silvered, the image will be formed at



- 1) 60 cm from left of AB 2) 30 cm from left of AB
3) 12 cm from left of AB 4) 60 cm from right of AB
33. In the arrangement show below, the image of the external object as seen by the observer is

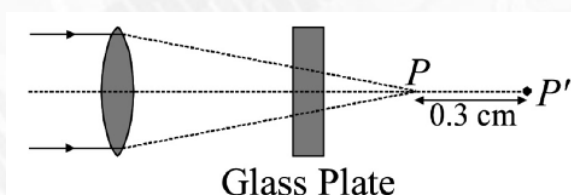


- 1) real and inverted 2) real and erect
3) virtual and inverted 4) virtual and erect

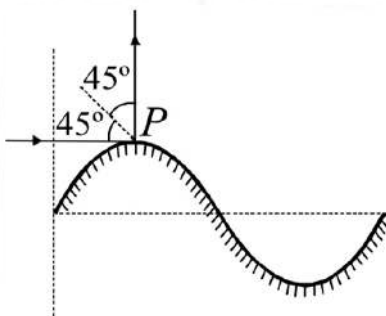




34. Assertion (A): Sound waves cannot propagate through vacuum but light can.
Reason (R) : Sound waves cannot be polarized.
- 1) Both (A) and (R) are correct and (R) is correct explanation of (R)
 - 2) Both (A) and (R) are correct and (R) is not correct explanation of (R)
 - 3) (A) is true (R) is false
 - 4) (A) is false (R) is true
35. A parallel beam of fast moving electrons is incident normally on a narrow slit. A screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statement is correct?
- 1) Diffraction pattern is not observed on the screen in the case of electron.
 - 2) The angular width of the central maximum of the diffraction pattern will increase.
 - 3) The angular width of the central maximum will decrease.
 - 4) The angular width of the central maximum will remains the same.
36. Rays from a lens are converging towards a point P, as shown in figure. How much thick glass plate having refractive index 1.6 must be located between the lens and point P, so that image will be formed at P'?



- 1) 0.8 cm
 - 2) 1.6 cm
 - 3) 5 cm
 - 4) 2.4 cm
37. A reflecting surface is given by $y = 2\sin x$. The reflecting surface is facing positive y-axis. What is the least value of coordinates of the point where a ray parallel to positive x-axis becomes parallel to positive y-axis after reflection?



- 1) $\left(\frac{\pi}{3}, \sqrt{3}\right)$
- 2) $\left(\frac{\pi}{2}, \sqrt{2}\right)$
- 3) $\left(\frac{\pi}{3}, \sqrt{2}\right)$
- 4) $\left(\frac{\pi}{4}, \sqrt{3}\right)$





38. Four light sources produce the following four waves

(i) $y_1 = a \sin(\omega t + \phi_1)$

(ii) $y_2 = a \sin 2\omega t$

(iii) $y_3 = a' \sin(\omega t + \phi_2)$

(iv) $y_4 = a' \sin(3\omega t + \phi)$

Superposition of which two waves give rise to sustained interference?

- 1) (i) and (ii) 2) (ii) and (iii) 3) (i) and (iii) 4) (iii) and (iv)

39. A source emits electromagnetic waves of wavelength 3 m. One beam reaches the observer directly and other after reflection from a water surface, travelling 1.5 m extra distance and with intensity reduced to $(1/4)$ as compared to intensity due to direct beam alone. The resultant intensity will be (as compared to intensity due to direct beam alone).

- 1) $(1/4)$ fold 2) $(3/4)$ fold 3) $(5/4)$ fold 4) $(9/4)$ fold

40. Statement-I : Corpuscular theory fails in explaining the velocities of light in air and water.

Statement-II : According to corpuscular theory, light should travel faster in denser media than in rarer media.

- 1) Both Statement-I and Statement-II are correct
2) Both Statement-I and Statement-II are incorrect
3) Statement-I is correct and Statement-II is incorrect
4) Statement-I is incorrect and Statement-II is correct

41. Statement I : The ratio of the amplitudes of electric and magnetic fields is constant and it is equal to velocity of the electromagnetic waves.

Statement II : It is because, $B_o = cE_o$

- 1) Both Statement-I and Statement-II are correct
2) Both Statement-I and Statement-II are incorrect
3) Statement-I is correct and Statement-II is incorrect
4) Statement-I is incorrect and Statement-II is correct

42. A coil has resistance 30 ohm and inductive reactance 20 ohm at 50 Hz frequency. If an ac source, of 200 volt, 100 Hz, is connected across the coil, the current in the coil will be

- 1) 2.0 A 2) 4.0 A 3) 8.0 A 4) $\frac{20}{\sqrt{13}}$ A



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43. Match the following:

Column I (Circuit Parameter/Condition)		Column II (Value/Principle)	
A.	Impedance of a series LCR circuit at resonance	P.	Conservation of Charge
B.	Phase difference (ϕ) between voltage and current in a purely resistive AC circuit	Q.	R (Resistance)
C.	Average power dissipated by an ideal inductor or capacitor in a full cycle of AC	R.	0°
D.	Principle on which Kirchhoff's Current Law (KCL) is based	S.	0 (Zero)

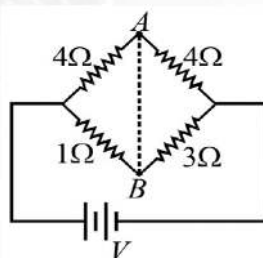
1) A-R; B-Q; C-S, D-P

2) A-Q; B-R; C-P, D-S

3) A-Q; B-R; C-S, D-P

4) A-P; B-S; C-R, D-Q

44. In the circuit shown, if a conducting wire is connected between points A and B, current in this wire will



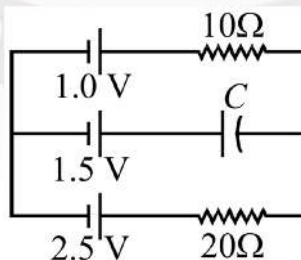
1) flow from B to A

2) flow from A to B

3) flow in the direction which will be decided by the value of V

4) be zero

45. In the following circuit diagram the potential difference across the plates of the capacitor C is



1) 2.5 volt

2) 1.5 volt

3) 1.0 volt

4) zero

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SECTION-II (NUMERICAL VALUE TYPE)

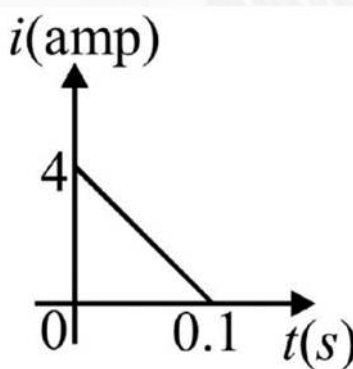
This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. In a cuboid of dimensions $2L \times 2L \times L$, a point charge q is placed at the centre of the face

having area $4L^2$. The flux through a face having area $2L^2$ is $\frac{q}{x\epsilon_0}$ then x is _____

47. In a coil of resistance 10Ω , the induced current developed by changing magnetic flux through it, is shown in the figure as a function of time. The magnitude of change in flux through the coil in weber is _____



48. The primary and secondary coils of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in webers, t is time in seconds and ϕ_0 is a constant, the output voltage (in volt) across the secondary coil is _____

49. A galvanometer has a coil of resistance 100 ohm and gives a full scale deflection for 30 mA current. If it is to work as a voltmeter of 30 volt range, the resistance (in ohm) required to be connected in series will be _____

50. The electric potential at a point (x, y) in the xy – plane is given by $V = -Kxy$. The magnitude of electric field intensity at a distance r from the origin varies as r^n . Value of n is _____

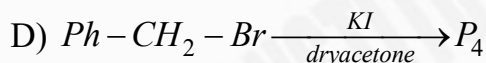
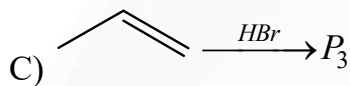
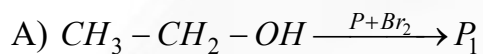


**CHEMISTRY****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. Correct SN^2 reactivity order of P_1, P_2, P_3 & P_4 is?



1) $P_4 > P_2 > P_3 > P_1$

2) $P_4 > P_2 > P_1 > P_3$

3) $P_1 > P_2 > P_3 > P_4$

4) $P_2 > P_4 > P_1 > P_3$

52. Statement-I : The haloalkanes are very less soluble in water

Statement-II : All methyl halides and ethyl chloride are gases at room temperature

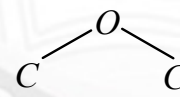
1) Both Statement-I and Statement-II are correct

2) Both Statement-I and Statement-II are incorrect

3) Statement-I is correct and Statement-II is incorrect

4) Statement-I is incorrect and Statement-II is correct

53. Statement-I : The bond angle of  in alcohols is slightly less than tetrahedral angle ($109^\circ 28'$)

Statement-II : In ethers  bond angle slightly greater than tetrahedral angle

1) Both Statement-I and Statement-II are correct

2) Both Statement-I and Statement-II are incorrect

3) Statement-I is correct and Statement-II is incorrect

4) Statement-I is incorrect and Statement-II is correct

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54. Assertion (A) : Aldehydes / Ketones having $\alpha-H$ atoms undergo aldol condensation / reaction
Reason (R) : Acidity of α -hydrogens of aldehydes and ketones is responsible for aldol reaction
- 1) Both (A) and (R) are correct and (R) is correct explanation of (A)
 - 2) Both (A) and (R) are correct and (R) is not correct explanation of (A)
 - 3) (A) is true (R) is false
 - 4) (A) is false (R) is true
55. Which of the following are essential α -amino acids and contain polar side chain?
A) Lysine B) Glutamic acid C) Tyrosine D) Isoleucine E) Cysteine
- 1) A only
 - 2) A & C
 - 3) C only
 - 4) B, D & E
56. Consider following reactions, in which of the following major organic product contain minimum number of π -bonds ?



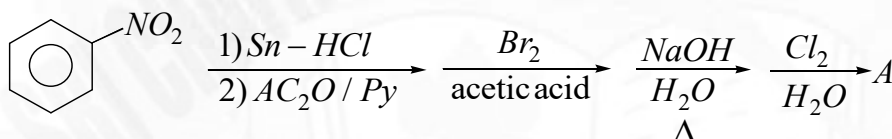
57.



DBE / DUS of x is

- 1) 4
- 2) 6
- 3) 8
- 4) 11

58.



A is

- 1)
- 2)
- 3)
- 4)

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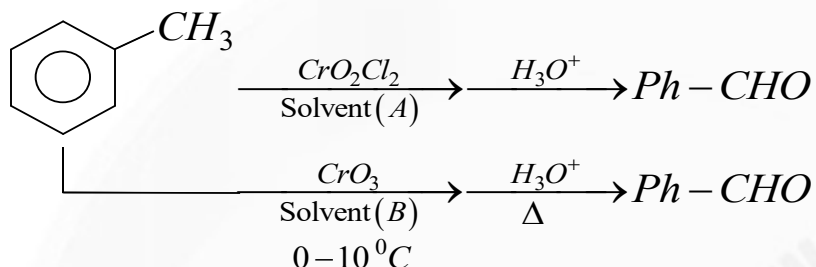
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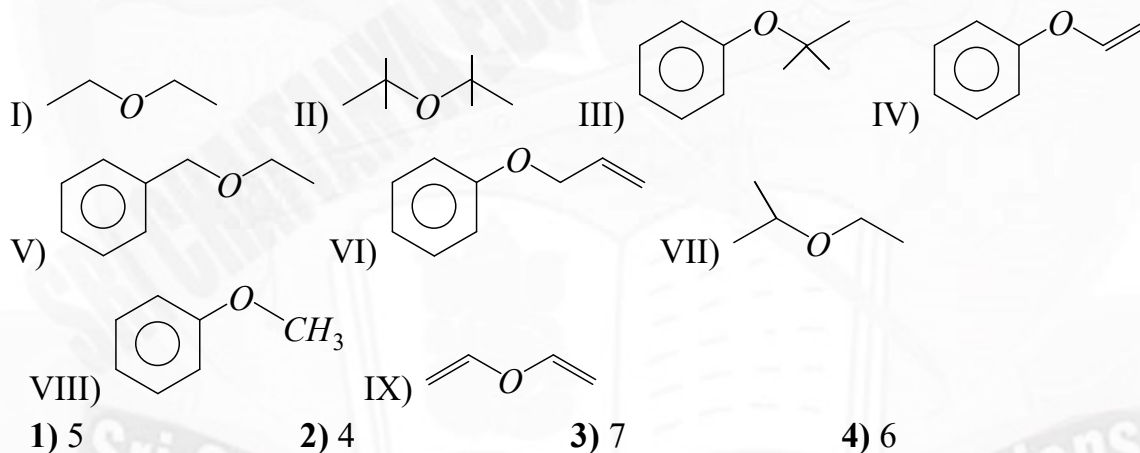
59. Correct acidic strength of following carboxylic acids is?
 a) 3-chloropropanoic acid b) 2-bromoethanoic acid
 c) benzoic acid d) Formic acid
 1) $b > a > c > d$ 2) $b > a > d > c$ 3) $b > d > a > c$ 4) $b > d > c > a$

60.

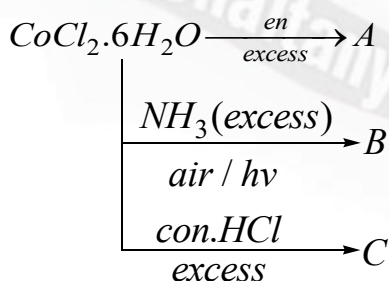


Solvent A and B respectively is ?

- 1) CS_2 & AC_2O 2) CS_2 & $\text{dil.H}_2\text{SO}_4$
 3) CS_2 & CS_2 4) AC_2O & $\text{dil.H}_2\text{SO}_4$
61. How many of the following can be prepared by williamson's ether synthesis in good yield



62.

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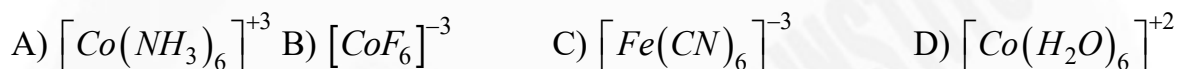


Correct statement(s) is/are ?

- i) A is formed as 2 isomeric complex where as B & C are formed as single complex
- ii) Co-ordination number of A, B, C are same
- iii) A, B, C have same geometry and same magnetic character
- iv) $\lambda_{\max}$ of C > A

- 1) i, ii & iv 2) i & iv only 3) i & ii only 4) i, iii & iv

63. Correct spin only magnetic momentum order of following complexes is



- 1) A = B > C > D 2) B > A > C > D
3) B > C > A > D 4) B > D > C > A

64. Assertion (A) : Cuprous ions are colourless where as Cu^{+2} is blue in the aqueous solution

Reason (R) : Cuprous ion has one unpaired d-electron while cupric ion does not have any unpaired d-electron

- 1) Both (A) and (R) are correct and (R) is correct explanation of (A)
2) Both (A) and (R) are correct and (R) is not correct explanation of (A)
3) (A) is true (R) is false
4) (A) is false (R) is true

65. Match the following

	Column-I (Physical property)		Column-II (Order)
A)	Ionisation enthalpy	P)	$HI > HBr > HCl > HF$
B)	Boiling point	Q)	$H_2O > H_2Te > H_2Se > H_2S$
C)	Melting point	R)	$BiH_3 > SbH_3 > NH_3 > PH_3$
D)	K_a values	S)	$F > N > O$

- 1) A-S; B-P; C-Q; D-R 2) A-S; B-Q; C-P; D-R
3) A-S; B-R; C-P; D-Q 4) A-S; B-R; C-Q; D-P

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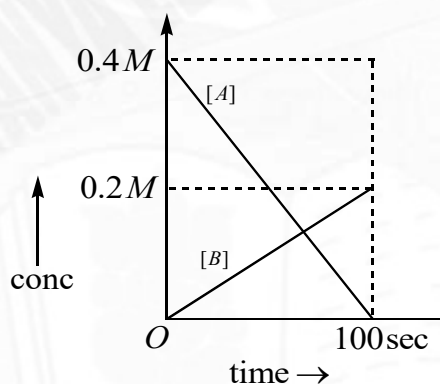


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66. Which element has atleast one 5d electrons in its +2 oxidation state?
 1) Gd 2) Ce 3) Eu 4) Tb
67. Incorrect statement is?
 1) All hexafluorides of G-16 are gaseous in nature
 2) Reducing property decreases from SO_2 to TeO_2
 3) All the hydrides of G-16 except water possess reducing property
 4) SO_2 is oxidizing and TeO_2 is reducing agents
68. A sample of sea water was found to contain 11.1% w/w of CaCl_2 , 9.5% w/w of MgCl_2 and 5.85% w/w of NaCl . Assuming complete ionization of salts, what will be the normal boiling point of sample. (K_b for water = 0.52 K·kg/mol, At. wt of $\text{Mg} = 24$, $\text{Ca} = 40$, $\text{Na} = 23$, $\text{Cl} = 35.5$, Normal boiling point of water = 100°C),
 1) 104.16°C 2) 105.66°C 3) 101.56°C 4) 102.12°C
69. For the reaction $x\text{A}_{(g)} \longrightarrow y\text{B}_{(g)} + \text{heat}$, which of the following statements are correct?
 (x, y are simplest whole numbers)



- I) Reaction is a zero order reaction
 II) Rate constant for the reaction is $4 \times 10^{-3} \text{ M/s}$
 III) If temperature of reaction mixture is increased, rate of reaction decreases
 IV) The reaction is a complex reaction
- 1) I, II, IV only 2) I, III, IV only
 3) I, IV only 4) All statements are correct





70. The vapour pressure of pure benzene and toluene are 800 torr and 600 torr respectively at 300 K. A liquid solution is formed by mixing 1 mole each of benzene & toluene. If the pressure over the solution is reduced slowly under isothermal condition, then mole fraction of benzene in the residue liquid, which has normal boiling point at 300 K is

1) 0.2

2) 0.57

3) 0.8

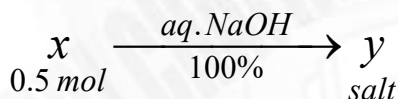
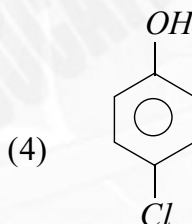
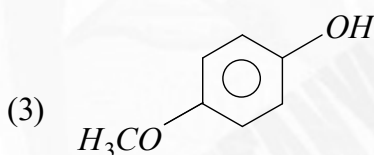
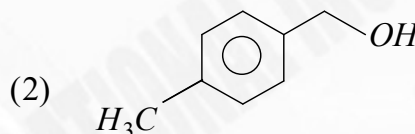
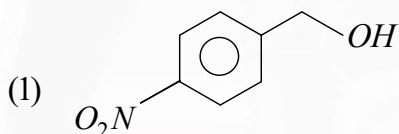
4) 0.43

SECTION-II (NUMERICAL VALUE TYPE)

This section contains 5 Numerical Value Type Questions. The Answer should be within 0 to 9999. If the Answer is in Decimal then round off to the Nearest Integer value (Example i.e. If answer is above 10 and less than 10.5 round off is 10 and If answer is from 10.5 and less than 11 round off is 11).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

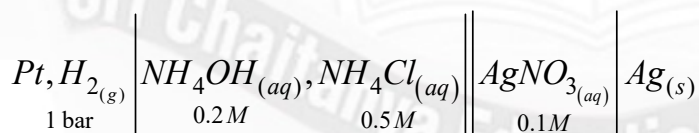
71. Among the following x is the most acidic compound



Weight of y formed is (in gms)?

(Consider At. Wt of $C = 12$ / $H = 1$ / $O = 16$; $Cl = 35$ / $Na = 23$ / $N = 14$)

72. Find the E_{cell} (in mV) for the following Galvanic cell :



Given : $E^0_{H^+/H_2} = 0.0V$, $E^0_{Ag^+/Ag} = +0.80V$

K_b of $NH_4OH = 10^{-5}$

Take $\frac{2.303 RT}{F} = 0.06V$, $\log 5 = 0.7$, $\log 3 = 0.48$, $\log 2 = 0.3$



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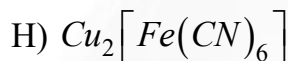
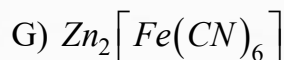
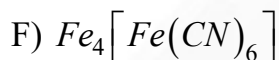
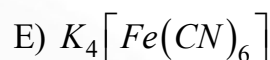
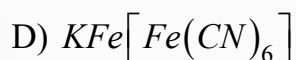
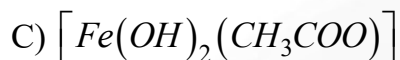
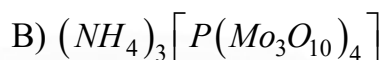
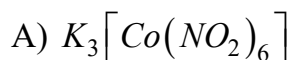
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73. How many of the following are yellow coloured

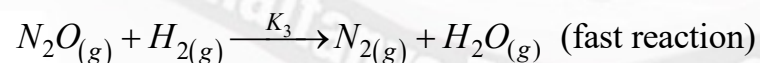
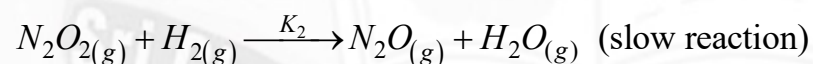
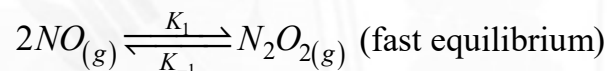


74. The specific conductance of 0.004 M acetic acid is $4 \times 10^{-5} S - cm^{-1}$ at 300 K temperature. The limiting molar conductivity of $CH_3COOH = 400 S - cm^2 / mol$, at the same temperature. Calculate the osmotic pressure (torr) of 0.001 M acetic acid at 300 K (Nearest integer)

$$\left[\text{Take } RT = 25 \frac{\text{atm-lit}}{\text{mol}}, 1 \text{ atm} = 760 \text{ torr} \right]$$

75. Consider the following reaction; $2H_{2(g)} + 2NO_{(g)} \longrightarrow N_{2(g)} + 2H_2O_{(g)}$

Which follows the mechanism given below:



The overall order of the reaction is _____

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